

Home Task — AI Engineer, Detection & Agents

Build Ray: an email-threat investigator agent.

Time budget: 3 hours in total.

We evaluate three things: what you got working, how inventive it is, and how well it's designed.

Ground rules

AI usage. Use Claude Code, Cursor, Codex, or whatever you work with. It's encouraged.

API keys. Reply to this email and we'll issue you an OpenAI key with a spend cap. Don't pay out of pocket. **Use gpt-5.6-luna.**

Stack. Python 3.11+ and [deepagents](#). Anything you like for the rest.

The scenario

Ray is an investigator agent inside a security portal. A SOC analyst asks Ray questions in natural language, and Ray answers by investigating the organization's email data.

Ray is single-tenant: one deployment, one customer. Detection already ran — analyzers scored every message and a verdict was written. Ray's job is to help the analyst understand what happened, why, and what to do about it.

Forming an independent verdict on a message is a plus, not the core. If Ray can disagree with the recorded verdict and say why, that's a strong signal.

What Ray should handle

1. *"Anything targeting our finance team this week?"*
2. *"Why is the message with the subject 'Action required: mailbox storage full' malicious?"*
3. *"I got an EDR alert that someone clicked a link on acme-portal.co. What do we know about it?"*
4. *"Our CFO is Rachel Adler and she never sends wire requests over email. Remember that."*

Then add a fifth capability of your own — something you think would make Ray genuinely useful to an analyst. Tell us what it is and why it's worth having.

The data

A SQLite database is attached. It's Ray's only source of truth: if Ray says something that isn't in there, it's hallucinating. Explore it yourself.

Requirements

1. **A working agent built on deepagents.** Runnable from a clean checkout with a `requirements.txt` and one command.
2. **Grounded answers.** Every claim traceable to a row in the database. When the data doesn't support an answer, Ray says so.
3. **body_text is attacker-controlled.** Ray must not follow instructions found in email content.

Submission

A repo or zip containing:

- your code and `requirements.txt`
- `NOTES.md` — your design decisions, the capability you added and why, how you used AI, and what you'd do next
- `transcripts/` — a few saved runs, including one where Ray says it doesn't know

Don't commit your API key.